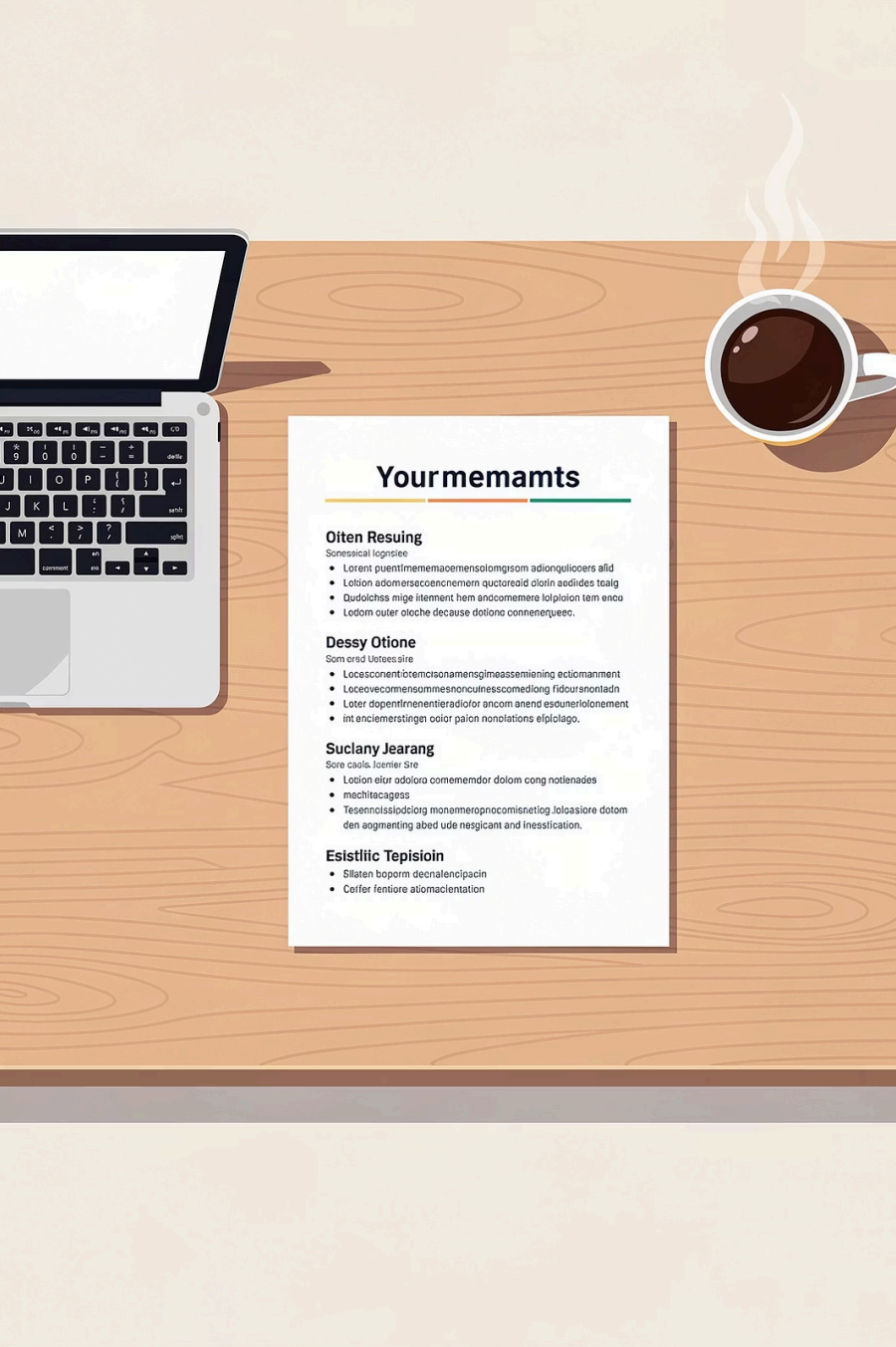


ResumeFit AI

An intelligent, multi-layer scoring engine that helps job seekers understand why their CV isn't passing ATS filters — and gives them concrete, actionable steps to fix it.

AI BOOTCAMP · BATCH 11

Bernard Lokasasmita



THE PROBLEM WE'RE SOLVING

Why Great Candidates Get Filtered Out Before a Human Ever Reads Their CV

Most job applications are screened by Applicant Tracking Systems (ATS) before reaching a hiring manager. Candidates with strong experience are routinely rejected — not because they're unqualified, but because their CV doesn't match the right keywords or formatting expectations.

ResumeFit AI bridges that gap. It analyzes a candidate's CV against a specific job description, surfaces exactly where the mismatch lies, and delivers a rewritten CV that is optimized to pass ATS screening — increasing the candidate's real chance of landing an interview.

Our Core Objective

- Diagnose why a CV fails ATS filters
- Score alignment across 4 dimensions
- Deliver concrete, prioritized fixes
- Generate a tailored CV rewrite instantly

Model Description: A 4-Layer Hybrid AI Scoring Engine

Layer 1 — TF-IDF Cosine Similarity

Statistical keyword overlap between CV and job description using term frequency-inverse document frequency vectors.

Layer 2 — Pinecone Vector Search

Semantic meaning captured via sentence embeddings stored and queried in Pinecone's vector database.

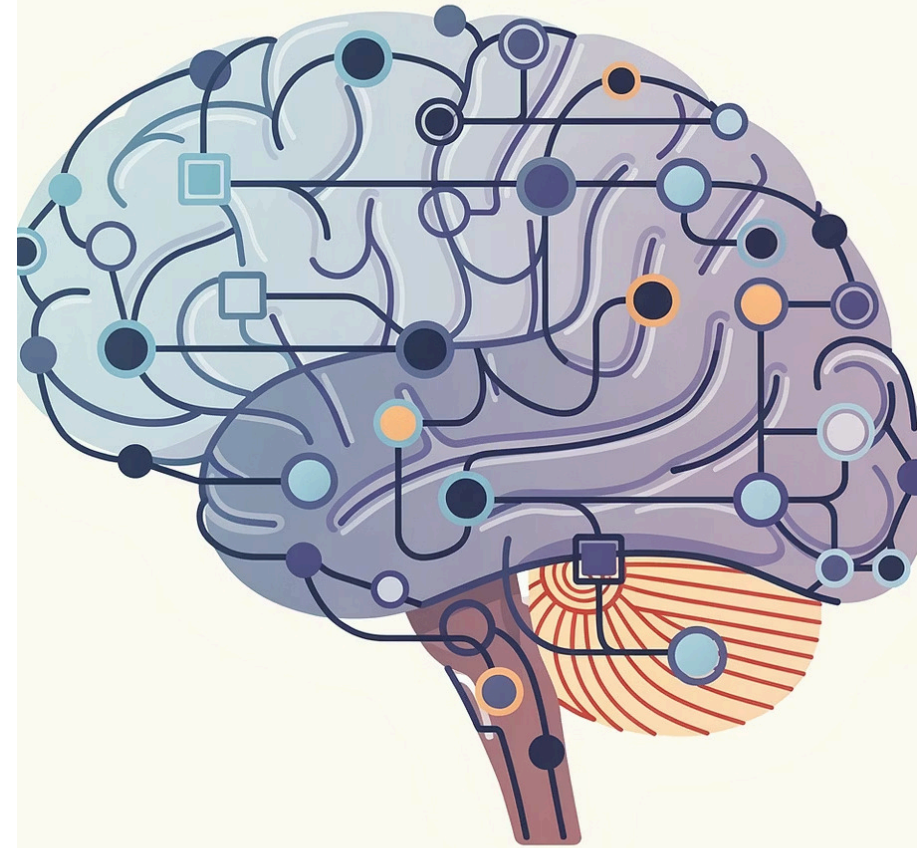
Layer 3 — Keyword Coverage

Exact keyword gap analysis — identifying which job description terms are present or missing in the CV.

Layer 4 — Skill Matching

Rule-based matching against a curated dictionary of 150+ technical and soft skills drawn from the JD.

On top of scoring: **LLM-powered 12-aspect CV evaluation**, impact language analysis, ATS formatting check, and smart CV rewrite via **Groq** (free tier) or **OpenAI**.



Scoring Metrics & Weighted Formula

Metric	Formula	Weight
TF-IDF Similarity	Cosine similarity of TF-IDF vectors	40%
Semantic Similarity	Pinecone cosine score (all-MiniLM-L6-v2)	30%
Keyword Coverage	$\frac{ \text{JD keywords} \cap \text{CV keywords} }{ \text{JD keywords} }$	20%
Skill Match	Matched skills / Total JD skills	10%

Final Score = $0.4 \times \text{TF-IDF} + 0.3 \times \text{Pinecone} + 0.2 \times \text{Keywords} + 0.1 \times \text{Skills}$

Score Guide

 **75–100%**

Excellent — strong ATS match

 **55–75%**

Good — minor gaps to address

 **35–55%**

Moderate — significant rewrites needed

 **0–35%**

Needs Work — consider full overhaul

Dataset & Data Pipeline

ResumeFit AI does not train a custom model. It leverages pre-trained embeddings and processes user data entirely at runtime.



Runtime Input Data

User-uploaded CV (PDF) and a pasted job description are the only inputs required. No personal data is stored between sessions.



Pre-Trained Embedding Model

all-MiniLM-L6-v2 from HuggingFace Sentence-Transformers — trained on over 1 billion sentence pairs for high-quality semantic representations.



Sample Resumes for Testing

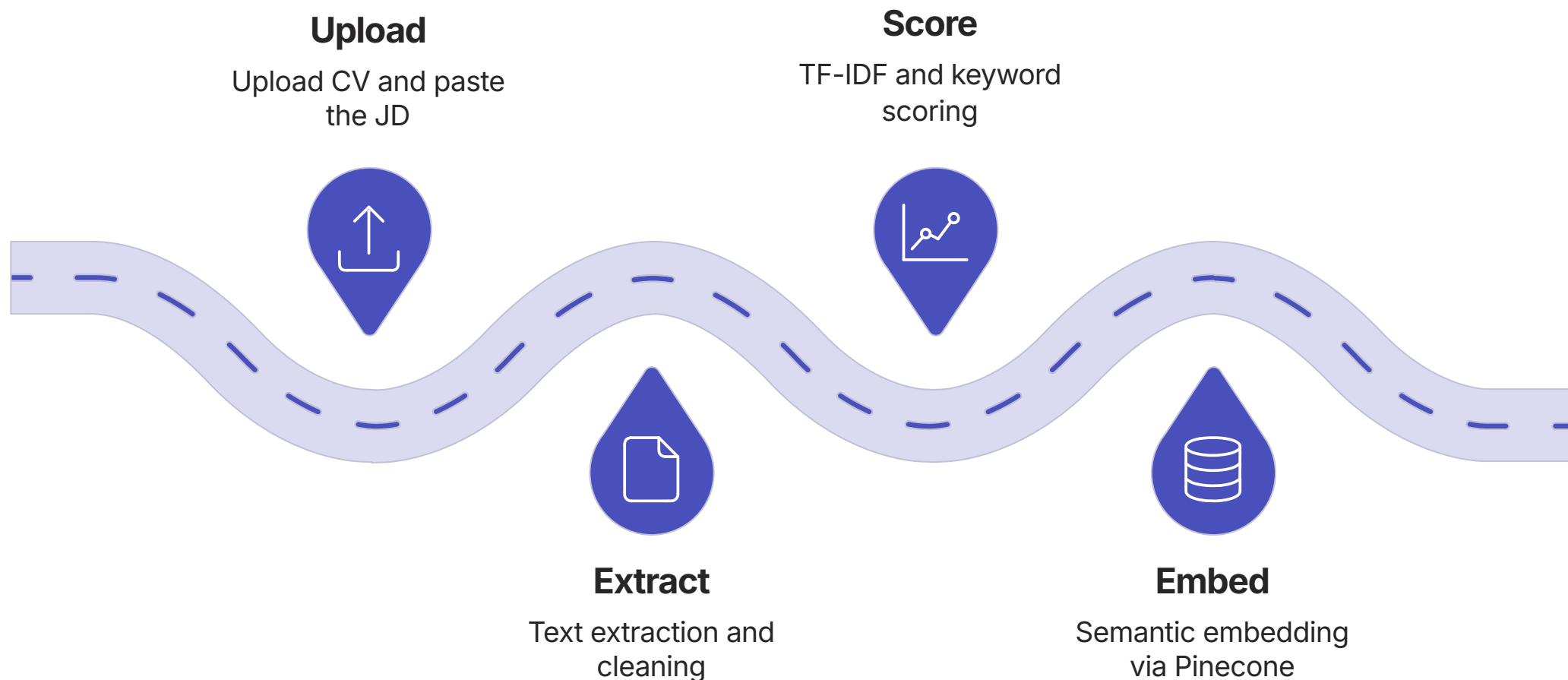
5 example CVs across marketing, engineering, and sales roles (in **Contoh Resumes/**) are included for demo and validation purposes.



Preprocessing Pipeline

PDF text extraction → lowercasing → regex cleaning → stopwords removal → vectorization. Handles most standard CV formats automatically.

End-to-End Methodology



The full pipeline runs end-to-end in under a minute. The LLM layer (Groq or OpenAI) handles nuanced language evaluation and produces a fully rewritten CV tailored to the specific job description — all within the same Colab notebook session.

Challenges, Solutions & Key Learnings

1

Pinecone Cold-Start Latency

Solution: Added spinner and real-time status messages during vector ingestion.

Learning: User feedback during processing matters as much as final accuracy.

2

Inconsistent LLM Output Formatting

Solution: Strict prompt engineering with exact output templates and labeled fields.

Learning: Structured prompts dramatically improve parsing reliability.

3

Image-Based PDF Parsing Failures

Solution: Built an ATS format checker that warns users when image-heavy CVs are detected.

Learning: Detection is as valuable as scoring.

4

Low-Match CV Force-Fitting

Solution: Dual strategy — tailor CV if score $\geq 60\%$, enhance overall if $< 60\%$.

Learning: AI should be honest, not just compliant.

5

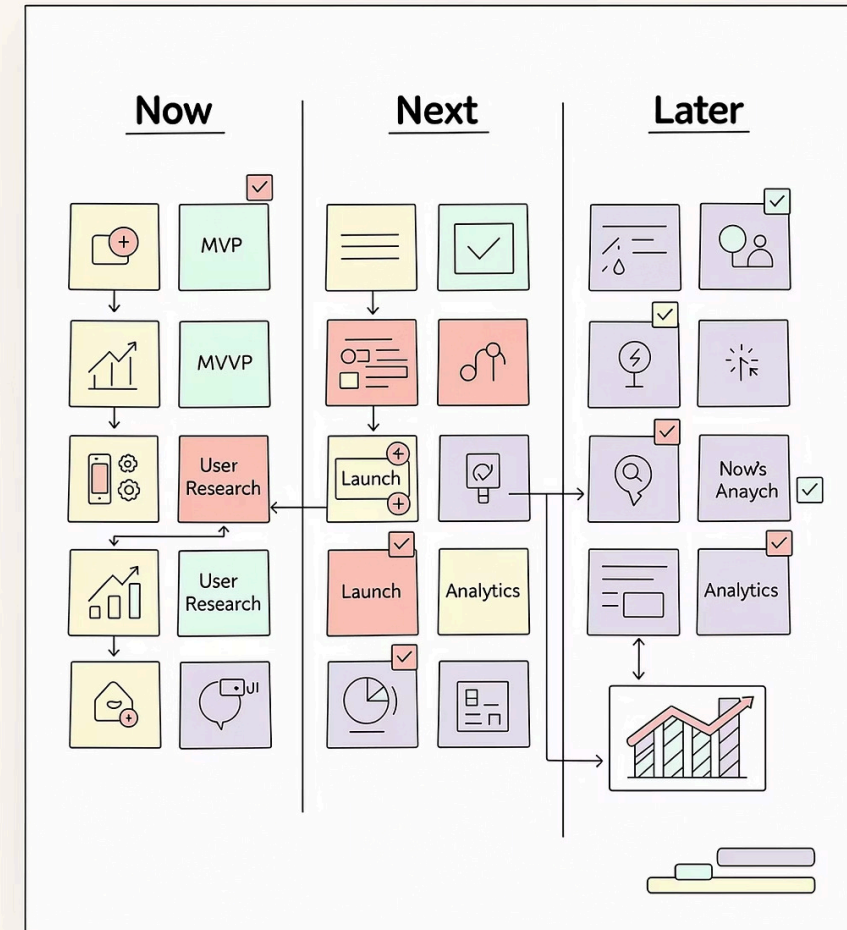
Groq Free-Tier Rate Limits

Solution: Built a full rule-based fallback engine for when API limits are hit.

Learning: Every AI feature needs a reliable non-AI fallback.

Future Improvements Roadmap

- 1 V2.0**
Multi-job comparison dashboard and full **Bahasa Indonesia** language support for local market expansion.
- 2 V3.0**
Job recommendation engine that suggests matching roles, plus automated CV template generation from scratch.
- 3 V4.0**
Analysis history to track CV improvements over time, plus full public web deployment beyond Google Colab.
- 4 V5.0**
Direct integration with **LinkedIn, JobStreet**, and other major job portals for one-click JD import and live match scoring.



Try It Now — Live Demo

Run on Google Colab

1 Open the Notebook

Launch `ResumeFit AI Bernard L v1.ipynb` in Google Colab.

2 Add Your Secrets

Set `NGROK_AUTH_TOKEN`, `GROQ_API_KEY`, and `PINECONE_API_KEY` in Colab Secrets.

3 Launch the App

Run both cells — a public URL is generated instantly via ngrok.

GitHub Repository

Full source code, sample resumes, and documentation are available on GitHub:

github.com/benloimAOG/AI-Boot-Camp-batch-11

Built With

Groq / OpenAI

LLM-powered rewrite

Pinecone

Vector search

HuggingFace

Sentence embeddings

✓ Designed, built, and deployed solo — end-to-end — as part of AI Bootcamp Batch 11.